

---

# SUPPLEMENTARY MATERIAL OF REVISITING IPS IN RECOMMENDATION MODELS: UNVEILING ITS IMPACT ON MODEL PERFORMANCE

---

## A Proof of Proposition 1

**Proposition 1.** (*Equivalence of Risk Minimizers*) Under the Assumptions 1–3, and assuming the uniqueness of the minimizer  $\theta^* = (\theta_z^*, \theta_f^*)$ ,

$$\arg \min_{\theta \in \Theta} \mathbb{E}[\widehat{R}_{ips}(\theta; \pi, \mathcal{D}_n)] = \arg \min_{\theta \in \Theta} \mathbb{E}[\widehat{R}_{naive}(\theta; \mathcal{D}_n)] = \theta^*.$$

*Proof.*

Let denote the risk function as

$$R(\theta; w, P_{Y,T,X}) = \int \frac{\mathbb{I}(t=1)}{w(x)} \mathbb{E}[\ell(Y, f(z(X; \theta_z); \theta_f)) | X = x] dP_{T,X}(t, x),$$

where  $w(x)$  is defined as the propensity score for the IPS risk, and set to 1 for the naive risk. Then,

$$\begin{aligned} R(\theta; w, P_{Y,T,X}) &= \int \frac{\mathbb{I}(t=1)}{w(x)} \mathbb{E}[-Y \log f(z(X; \theta_z); \theta_f) \\ &\quad - (1 - Y) \log(1 - f(z(X; \theta_z); \theta_f)) | X = x] dP_{T,X}(t, x) \\ &= \int \frac{\mathbb{I}(t=1)}{w(x)} \{ -\mathbb{P}(Y = 1 | T = 1, X = x) \log f(z(x; \theta_z); \theta_f) \\ &\quad - \mathbb{P}(Y = 0 | T = 1, X = x) \log(1 - f(z(x; \theta_z); \theta_f)) \} dP_{T,X}(t, x) \end{aligned}$$

By differentiating  $R(\theta; w, P_{Y,T,X})$  with respect to  $\theta$  to find the minimizer,

$$\begin{aligned} \frac{\partial}{\partial \theta} R(\theta; w, P_{Y,T,X}) &= \int \frac{\mathbb{I}(t=1)}{w(x)} \left\{ -\frac{\mathbb{P}(Y = 1 | T = 1, X = x)}{f(z(x; \theta_z); \theta_f)} \right. \\ &\quad \left. + \frac{\mathbb{P}(Y = 0 | T = 1, X = x)}{1 - f(z(x; \theta_z); \theta_f)} \right\} \frac{\partial}{\partial \theta} f(z(x; \theta_z); \theta_f) dP_{T,X}(t, x) \end{aligned}$$

The gradient of the risk function with respect to  $\theta$  at  $\theta^* = (\theta_z^*, \theta_f^*)$  satisfies,

$$\begin{aligned} \frac{\partial}{\partial \theta} R(\theta; w, P_{Y,T,X}) &= \int \frac{\mathbb{I}(t=1)}{w(x)} \left\{ -\frac{\mathbb{P}(Y = 1 | T = 1, X = x)}{f(z(x; \theta_z^*); \theta_f^*)} \right. \\ &\quad \left. + \frac{\mathbb{P}(Y = 0 | T = 1, X = x)}{1 - f(z(x; \theta_z^*); \theta_f^*)} \right\} \frac{\partial}{\partial \theta} f(z(x; \theta_z^*); \theta_f^*) dP_{T,X}(t, x) \\ &= 0. \end{aligned}$$

This result holds regardless of the specific form of the weighting function  $w(x)$ , since at the true parameter  $\theta^*$ , the conditional probability  $\mathbb{P}(Y = 1 | T = 1, x)$  exactly matches the predicted probability  $f(z(x; \theta_z^*); \theta_f^*)$ . Consequently, both IPS and naive estimators share the same minimizer  $\theta^*$ , indicating that the optimal parameters are invariant to the weighting scheme when the model is correctly specified.

■

## B Risk Functions of Baselines

The following describes the composition of the loss functions for the methods used as baselines in our experiments. The tilde symbol ( $\sim$ ) indicates fixed parameters.

**IPS** [3]: A single-task learning method using IPS risk estimator.

$$\min_{\varphi_z, \varphi_\pi} \widehat{R}_{select}(\varphi_z, \varphi_\pi) = \frac{1}{n} \sum_{i=1}^n \ell(t_i, \pi(z(x_i; \varphi_z); \varphi_\pi)),$$

then,

$$\begin{aligned} \min_{\boldsymbol{\theta}} \widehat{R}_{ips}(\boldsymbol{\theta}; \tilde{\varphi}_z, \tilde{\varphi}_\pi) &= \frac{1}{n} \sum_{i=1}^n \left\{ \frac{\mathbb{I}(t_i = 1)}{\pi(z(x_i; \tilde{\varphi}_z); \tilde{\varphi}_\pi)} \ell(y_i, f(t_i, z(x_i; \boldsymbol{\theta}_z); \boldsymbol{\theta}_f)) \right. \\ &\quad \left. + \frac{\mathbb{I}(t_i = 0)}{1 - \pi(z(x_i; \tilde{\varphi}_z); \tilde{\varphi}_\pi)} \ell(y_i, f(t_i, z(x_i; \boldsymbol{\theta}_z); \boldsymbol{\theta}_f)) \right\}, \end{aligned}$$

where  $(y_i, x_i, t_i) \in \mathcal{D}_n$  and  $\boldsymbol{\theta} = (\boldsymbol{\theta}_z, \boldsymbol{\theta}_f)$ .

**AKB-IPS** [1]: Improved IPS method that adaptively selects the most impactful kernel functions in RKHS to achieve better causal balancing, reducing estimation bias while maintaining low variance.

$$\begin{aligned} \min \widehat{R}_{akbips}(\boldsymbol{\theta}; \hat{\omega}) &= \frac{1}{n} \sum_{i=1}^n \left\{ \frac{\mathbb{I}(t_i = 1)}{\hat{\omega}_i} \ell(y_i, f(z(x_i; \boldsymbol{\theta}_z); \boldsymbol{\theta}_f)) \right. \\ &\quad \left. + \frac{\mathbb{I}(t_i = 0)}{1 - \hat{\omega}_i} \ell(y_i, f(z(x_i; \boldsymbol{\theta}_z); \boldsymbol{\theta}_f)) \right\}, \end{aligned}$$

where  $(y_i, t_i, x_i) \in \mathcal{D}_n$ ,  $\boldsymbol{\theta} = (\boldsymbol{\theta}_z, \boldsymbol{\theta}_f)$ , and  $\hat{\omega} = (\hat{\omega}_1, \dots, \hat{\omega}_n)$  is the balancing weights learned via adaptive kernel balancing method. Please refer to original paper [1] for details.

**ESMM** [2]: Multi-task learning that simultaneously learns the propensity score  $\mathbb{P}(T = 1|x)$  and the joint distribution  $\mathbb{P}(Y = y, T = t|x) = \mathbb{P}(Y = y|T = t, x)\mathbb{P}(T = t|x)$ .

$$\min_{\boldsymbol{\theta}, \theta_\pi} \widehat{R}_{ctcvt}(\boldsymbol{\theta}, \theta_\pi) + \lambda_1 \widehat{R}_{select}(\boldsymbol{\theta}_z, \theta_\pi),$$

where  $(y_i, x_i, t_i) \in \mathcal{D}_n$ ,  $\boldsymbol{\theta} = (\boldsymbol{\theta}_z, \boldsymbol{\theta}_f)$ , and

$$\begin{aligned} \widehat{R}_{ctcvt}(\boldsymbol{\theta}, \theta_\pi) &= \frac{1}{n} \sum_{i=1}^n \{ \mathbb{I}(t_i = 1) \ell(y_i t_i, f(z(x_i; \boldsymbol{\theta}_z); \boldsymbol{\theta}_f)) \pi(z(x_i; \boldsymbol{\theta}_z); \theta_\pi) \\ &\quad + \mathbb{I}(t_i = 0) \ell(y_i (1 - t_i), f(z(x_i; \boldsymbol{\theta}_z); \boldsymbol{\theta}_f)) (1 - \pi(z(x_i; \boldsymbol{\theta}_z); \theta_\pi)) \}. \end{aligned}$$

## References

- [1] Haoxuan Li, Chunyuan Zheng, Yanghao Xiao, Peng Wu, Zhi Geng, Xu Chen, and Peng Cui. Debiased collaborative filtering with kernel-based causal balancing. *ArXiv*, abs/2404.19596, 2024.
- [2] Xiao Ma, Liqin Zhao, Guan Huang, Zhi Wang, Zelin Hu, Xiaoqiang Zhu, and Kun Gai. Entire space multi-task model: An effective approach for estimating post-click conversion rate. In *The 41st International ACM SIGIR Conference on Research & Development in Information Retrieval*, SIGIR '18, page 1137–1140, New York, NY, USA, 2018. Association for Computing Machinery.
- [3] Tobias Schnabel, Adith Swaminathan, Ashudeep Singh, Navin Chandak, and Thorsten Joachims. Recommendations as treatments: debiasing learning and evaluation. In *Proceedings of the 33rd International Conference on International Conference on Machine Learning - Volume 48*, ICML'16, page 1670–1679. JMLR.org, 2016.